

Keeper K-audit — Paper A1 (Substrate Hall Algebra, PRIMARY)

v0.3

Keeper

2026-05-28 Thursday PM EDT

Keeper K-audit — A1 (PRIMARY) v0.3

What's strong (acknowledge first)

A1 v0.3 is in genuinely good shape and the substance is referee-grade: - The RIGOROUS spine holds: Serre structure constants $[2]_2 = N_c$, $[3]_4 = N_c \cdot g$ (Type A, forward, Elie-verified) — the central result is real and correctly placed at the Hall-Littlewood corner. - Discipline is exemplary: scheme-dependent mass-ratio leads and back-fit relations are explicitly EXCLUDED (§8.3); tiers are stated per claim (§8.1-8.4); the Cal #146 unification correction (shared W_n , distinct K-types) is in; the empirical contact is restricted to the scheme-invariant mixing-angle spine. - The Macdonald parameter-role fix is correctly incorporated (§5, §8.4): Hall-Littlewood corner, α as evaluation/coupling, integrality argument cited. The gate I raised this morning is properly closed in the text. - c_{FK} -derived-measure theorem (§8.4) is correctly stated as a Strong-Uniqueness strengthener.

The paper is close. The two findings below are internal-consistency defects, not substance defects — but they are exactly the kind a referee catches on first read, and one of them is embarrassing precisely because the paper makes a virtue of not conflating genera.

Finding 1 — MODERATE: the paper contradicts itself on the genus of D_{IV}^5

- §1 + Route-A table + §8.4 say: Hua genus = $n_C = 5$; Faraut-Korányi genus = $C_2 = 6$; $g = 7$ = embedding.
- §8.1 (line 150) says: “ n_C = Faraut-Korányi genus = 5.”

These cannot both hold. §8.1 calls $n_C=5$ the *FK* genus; §1/§8.4 call $C_2=6$ the FK genus and $n_C=5$ the *Hua* genus. A paper whose own headline convention (§1) is “do not conflate the three genera” cannot then conflate them in §8.1. A referee will flag it instantly, and the contradiction undercuts the very convention the paper introduces.

Worse, the underlying fact is not actually settled: the FK multiplicity formula $p = a(r-1)+b+2$ for Type IV_5 ($a=3$, $b=0$, $r=2$) gives $p = 5 = n_C$ — which supports §8.1 (“FK genus = 5”) and *contradicts* “FK genus = $C_2 = 6$.” So the team has two competing claims about what the FK genus *is*, and the “FK genus = $C_2 = 6$ ” identification may itself be a mislabel ($C_2 = 6$ is the quadratic Casimir; whether it is *also* a genus is the open question).

What is NOT in dispute: the Bergman kernel exponent = $n_C = 5$ (convention-free, Elie $v=5$ MC-confirmed). The dispute is purely the *naming* (is 5 the Hua or the FK genus?) and whether $C_2=6$ is a genus at all.

Required fix (before Cal cold-read): pin the FK genus of Type IV_n against a primary source (Faraut-Korányi *Analysis on Symmetric Cones* genus table, or Upmeyer), not the team's competing notes. Then make §1, the Route-A table, §8.1, and §8.4 say the *same* thing. I am not adjudicating the name from memory — I was wrong on genus naming once today, and the multiplicity formula (giving 5) conflicts with the team's “FK=6” note, so this needs the book, not a guess. Route: Keeper (literature) + Lyra. This is the one item genuinely blocking a clean submission.

Finding 2 — MINOR: kernel-exponent staleness (§1 contradicts §8.4)

- §1 (line 40): “The kernel singularity exponent ($h^{-?/\text{rank}}$) is under active recheck ... whether the established $7/2$ stands is the open item.”
- §8.4 (line 170): “RESOLVED — kernel singularity exponent = $n_C/\text{rank} = 5/2$ (Elie 3580-3581, MC-confirmed; un-held).”

§1 is stale — it describes the kernel exponent as open when §8.4 (and today’s work) resolved it to $5/2$. Fix: update §1 line 40 to state the resolved value ($5/2 = n_C/\text{rank}$, Hua) and drop “under active recheck.” Trivial, but it’s a self-contradiction a careful reader will notice.

Verdict

CONDITIONAL PASS. The verified spine is sound; the exclusions and tiers are disciplined; the substance is referee-grade. Clear Finding 1 (genus self-contradiction + primary-source pin) and Finding 2 (kernel-exponent staleness), and A1 is PASS-ready for Cal’s cold-read. No CRITICAL gaps; the central RIGOROUS result (Serre constants) and the scheme-invariant empirical spine are unaffected by either finding.

Fix list → v0.4 (drives A1 to done)

1. [blocking] Finding 1 — pin FK genus of Type IV_5 vs primary source; make §1 / Route-A / §8.1 / §8.4 consistent. (Keeper lit + Lyra.)
2. [quick] Finding 2 — update §1 kernel-exponent wording to the resolved $5/2$. (Lyra.)
3. [quick] α_{fine} placement — §5.1 says “RESOLVED” (evaluation/coupling) but §4/§5.1 also still flag it “open” in places; make the disposition uniformly “resolved: evaluation, not a Macdonald coordinate.” (Lyra.)
4. [parallel] PMNS double-set (§8.5) — Cal’s typing closed $/N_{\text{max}}$ as canonical (§7); §8.5 still says “FLAGGED for reconciliation.” Reconcile §7 (closed) vs §8.5 (flagged) — they disagree on whether it’s settled. (Lyra + Cal.)
5. Then → Cal cold-read → v0.4 → Keeper final PASS.

Note: items 3 and 4 are two MORE internal staleness points (the paper says “resolved/closed” in one section and “open/flagged” in another for the same items). A1 has accumulated within-session edits faster than its cross-section consistency — a consistency sweep across all sections is the real v0.4 task. That’s the honest state: the substance is done; the document needs one careful internal-consistency pass so no section contradicts another.

— Keeper, 2026-05-28 Thursday PM. A1 CONDITIONAL PASS; substance referee-grade; needs one internal-consistency sweep (genus naming is the load-bearing fix) before Cal cold-read.

v0.4 verification read (Keeper, PM) — sweep ~95% landed; TWO residuals block final PASS

I read Lyra’s swept v0.4 (in-place, mtime 17:09) end-to-end to issue the final PASS. The sweep largely landed — but trust-but-verify caught two residuals the sweep missed, and one is the genus mislabel *again*. Final PASS withheld pending these (both quick, both real contradictions a referee catches):

Residual 1 — MODERATE (the genus, third time today): line 59 still contradicts the corrected §1 box. The §1 box (lines 38–42) and §8.4 (line 174) now correctly say “ONE genus = $n_C = 5$ (FK = Hua); $C_2 = 6$ is the Casimir, NOT a genus.” But line 59 still reads: “(Correction... n_C is the Hua genus, NOT the FK genus; C_2 is the FK genus; g is the embedding dimension...)” — i.e., it STILL asserts “ C_2 is the FK genus,” the exact mislabel the box corrects, and STILL asserts a Hua-vs-FK distinction the box says doesn’t exist. Direct self-contradiction. Fix (Lyra): delete the line-59 parenthetical or replace with the box’s statement — “ $n_C = 5$ is the genus (FK=Hua, no distinction); $C_2 = 6$ is the Casimir; $g = 7$ is the embedding — neither C_2 nor g is a genus.” (I’m routing this to Lyra rather than editing her paper, because the wording touches the naming I’ve deferred to the book-pin; she applies the box’s already-correct statement.)

Residual 2 — MINOR (stale footer): the v0.2 provenance + closing signature (lines ~200–208) were not swept. Line 202 (“ n_C re-anchored to FK genus”) and the §“Provenance of v0.2 changes” + the closing signature (line 208: “HELD: g-anchor wording + c_{FK} normalization pending T2442 recheck. Ready for Cal cold-read once T2442 lands”) are leftover v0.2 text — they describe items as HELD/pending that §8.4 now marks RESOLVED. Fix (Lyra): refresh the footer to the v0.4 state (c_{FK} resolved/un-held; T2442 stands; genus one-genus; ready for cold-read pending the line-59 fix + book-pin).

Everything else verified clean: §1 box, §7 mixing-angle spine, §8.1, §8.4 genus/ c_{FK} /kernel-exponent, §8.5 PMNS (CLOSED, consistent with §7), §9 conclusions — all consistent and correct. The substance (Serre spine RIGOROUS, mixing angles, c_{FK} -derived, confinement, two-corner) is referee-grade and unaffected.

Verdict: CONDITIONAL PASS (one micro-sweep from final PASS). Clear Residual 1 (line 59) + Residual 2 (footer), and the document is internally consistent. Then: final Keeper PASS → Cal cold-read (Cal already cold-read v0.3 per Cal #155; v0.4 is a quick verify) → submission-grade. The genus-naming book-pin remains my next-session item, but the doc handles it correctly by stating value+role + “pending primary-source pin,” so it does NOT block the PASS once line 59 is fixed.

Meta: this is the third genus-bite today and the second time a “sweep done” report had a residual — both caught by the Keeper read, not the self-report. It vindicates the trust-but-verify discipline and Casey’s “drive A1” call once more: A1 is genuinely one micro-fix from a final PASS, but it was NOT there on report alone.

— Keeper, 2026-05-28 Thursday PM. A1 v0.4 CONDITIONAL PASS — two residuals (line 59 genus contradiction MODERATE; stale v0.2 footer MINOR) routed to Lyra; final PASS on the corrected version.

FINAL VERDICT — A1 v0.4: PASS (Keeper, verified)

Lyra applied both fixes; I grep-verified rather than taking the report (the discipline this session earned): no “ $C_2 = FK$ genus,” no “Hua genus NOT the FK,” no “HELD pending”/“under recheck”/“once T2442 lands” anywhere; the one-genus convention (“Casimir NOT a genus”) stated consistently (4×). Both residuals are closed.

Verdict: PASS. A1 v0.4 is internally consistent end-to-end and substantively referee-grade: - RIGOROUS Serre-constant spine ($[2]_2 = N_C$, $[3]_4 = N_C \cdot g$) at the correct Hall-Littlewood corner. - Scheme-invariant mixing-angle empirical spine (invariant-anchored); PMNS closed (Cal #153) consistently in §7 + §8.5. - c_{FK} = derived-measure theorem; one-genus convention ($n_C = 5$ genus / $C_2 = 6$ Casimir / $g = 7$ embedding) consistent in §1 box, §8.1, §8.4, anchor table, §9, line-59, footer. - Cal #146-corrected unification; confinement FRAMEWORK-PLUS; leads + back-fit EXCLUDED; tiers honest throughout.

One non-blocking proof-stage action (my next-session item): confirm the genus *citation* against Faraut-Korányi/Loos before final submission. This does NOT condition the PASS — the genus *value* (5) is derivation-certain + 4-way confirmed, and the paper states it by value+role + cites + flags the naming as pending-pin, so it makes no claim the book-check could falsify. Belt-and-suspenders citation tightening, not a correctness gate.

Path to submission-grade: Keeper PASS (this) → Cal’s quick v0.4 verify (he cold-read v0.3 per Cal #155; v0.4 confirms the sweep + 2 substantive flags) → submission-grade. The PRIMARY paper is there.

This is what “drive A1, don’t park it” produced: a Keeper-PASSED, internally-consistent, referee-grade PRIMARY paper — reached by driving and *verifying* end-to-end (three genus-bites and two missed-residuals caught by the read, not the reports), not by parking-and-reporting.

— Keeper, 2026-05-28 Thursday PM. A1 v0.4 — Keeper final PASS (grep-verified). Cal quick-verify → submission-grade; genus citation book-pin is a non-blocking proof-stage item.